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INNER PRODUCT

Def^n:- Let V be a real vector space. A mapping $\langle \cdot, \cdot \rangle$, $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$ is said to be an inner product on V if it satisfies the following axioms

I1 Linear property

$$\langle \alpha u_1 + \beta u_2, v \rangle = \alpha \langle u_1, v \rangle + \beta \langle u_2, v \rangle$$

I2 Symmetric Property

$$\langle u, v \rangle = \langle v, u \rangle$$

I3 Positive definite property

$\langle u, u \rangle \geq 0$ and $\langle u, u \rangle = 0$ iff $u = 0$

↳ The vector space V with innerproduct is called innerproduct space.

Ex:- $\mathbb{R}^n \to$ Eucleidian space

$\langle \cdot, \cdot \rangle : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ st $\langle u, v \rangle = u \cdot v$

Check it is inner product or not.

Sol^n:- Let $u = (x_1, x_2, \ldots, x_n)$

$v = (y_1, y_2, \ldots, y_n)$

$$\langle u, v \rangle = u \cdot v = (x_1, x_2, \ldots, x_n)(y_1, y_2, \ldots, y_n)$$

$$= \sum_{i=1}^{n} x_i y_i$$

I1

$u_1 = (x_1, x_2, \ldots, x_n)$

$u_2 = (y_1, y_2, \ldots, y_n)$

$v = (z_1, z_2, \ldots, z_n)$

$$\langle \alpha u_1 + \beta u_2, v \rangle = \langle \alpha(x_1, \ldots, x_n) + \beta(y_1, \ldots, y_n), (z_1, z_2, \ldots, z_n) \rangle$$

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